Demo: Table Design - Compression

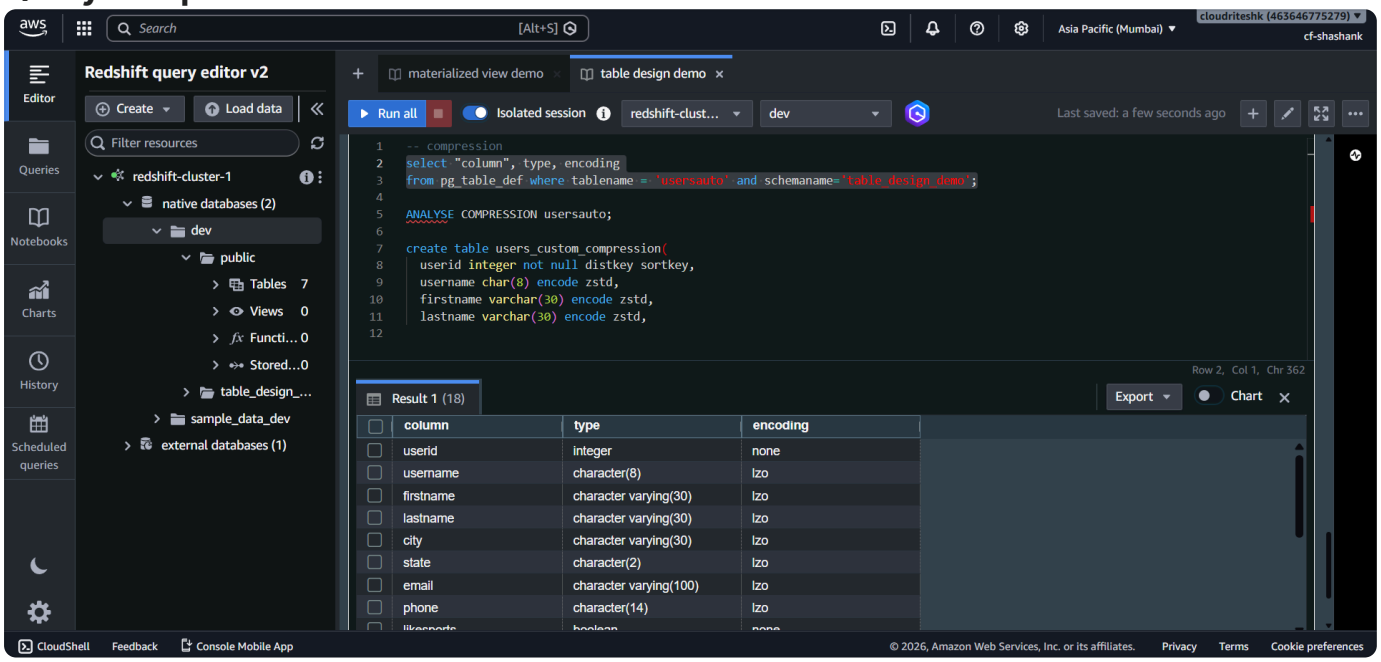
Summary

Checked default Redshift compression encoding using `pg_table_def`, analyzed recommended encodings with `ANALYZE COMPRESSION`, and created a custom table using `ZSTD` compression for selected columns to optimize storage and performance.

Check Existing Compression Encoding

Target table: `usersauto`

Query Compression Information




```
SELECT tablename,s
       "column",
       encoding
FROM pg_table_def
WHERE tablename = 'usersauto';
```

Observations

Dist Key / Sort Key `userid` has: `No Encoding`

Reason

Redshift does not compress:

- Distribution keys
- Sort keys

Default Character Encoding

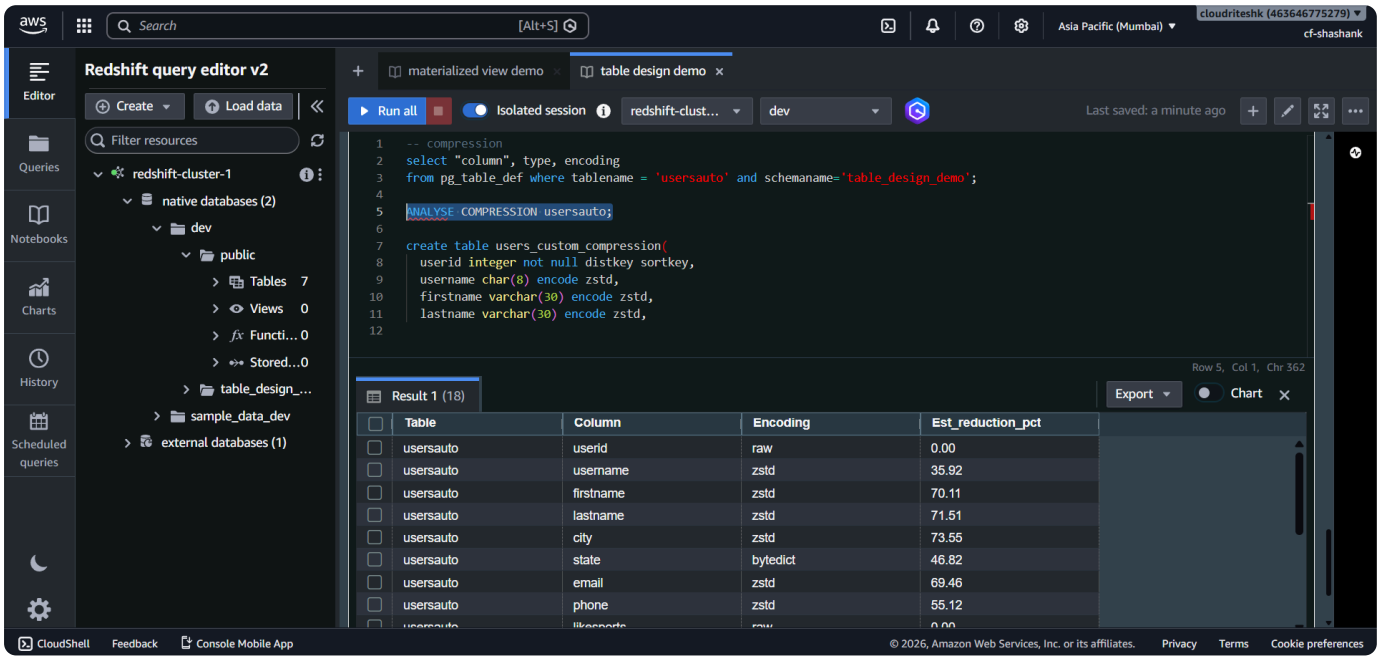
Columns like:

- username
- firstname
- lastname
- phone

used: `LZO` encoding by default.

Analyze Compression Recommendations

ANALYZE COMPRESSION Command




```
ANALYZE COMPRESSION usersauto;
```

Output Includes

Table column, Recommended encoding, Estimated storage reduction

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Recommendation Observed

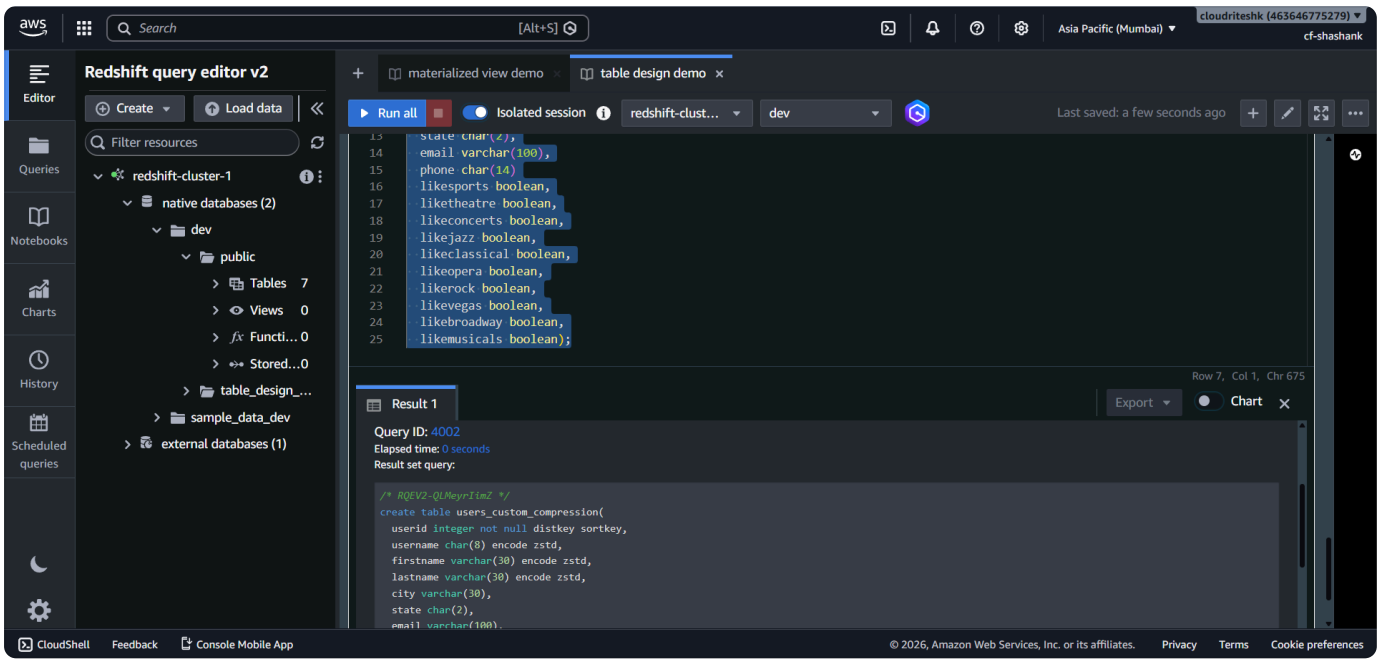
For columns like:

- firstname
- lastname
- username

Redshift recommended: `ZSTD` encoding.

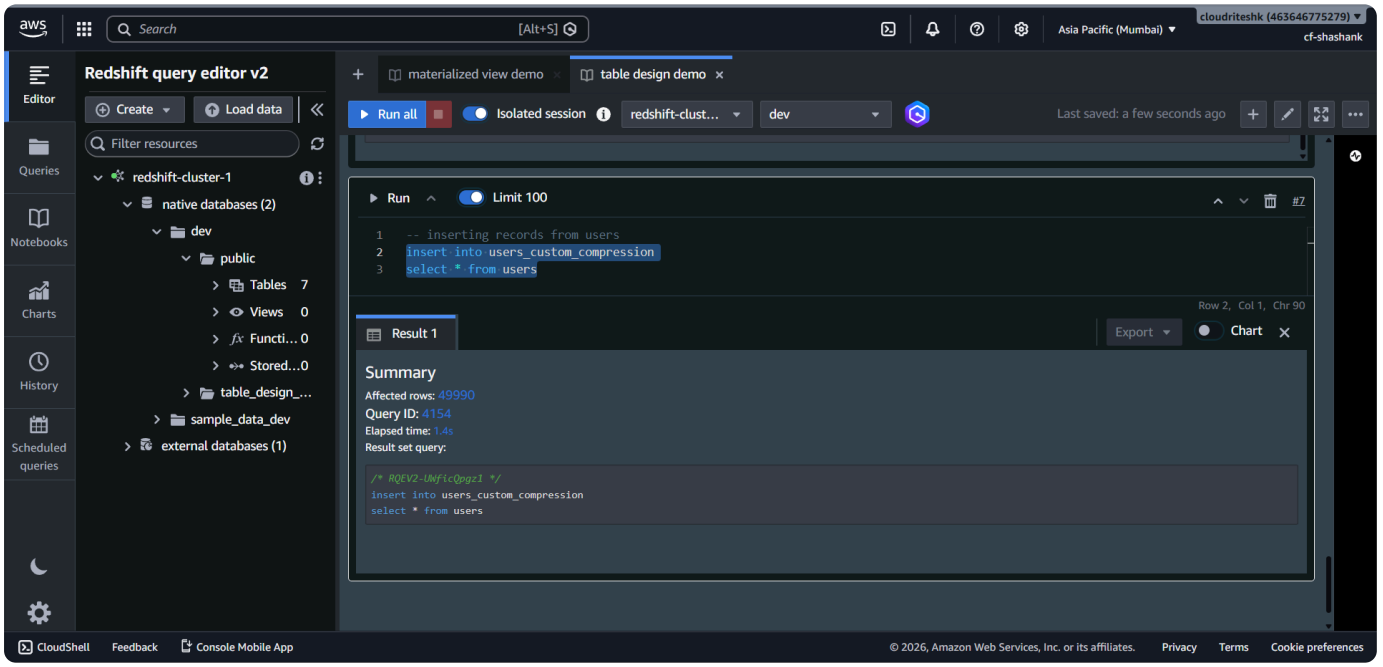
Create Table with Custom Compression

Create Table




```
CREATE TABLE users_custom_encoding (
    userid INTEGER,s
    username VARCHAR ENCODE zstd,
    firstname VARCHAR ENCODE zstd,
    lastname VARCHAR ENCODE zstd,
    city VARCHAR
);
```

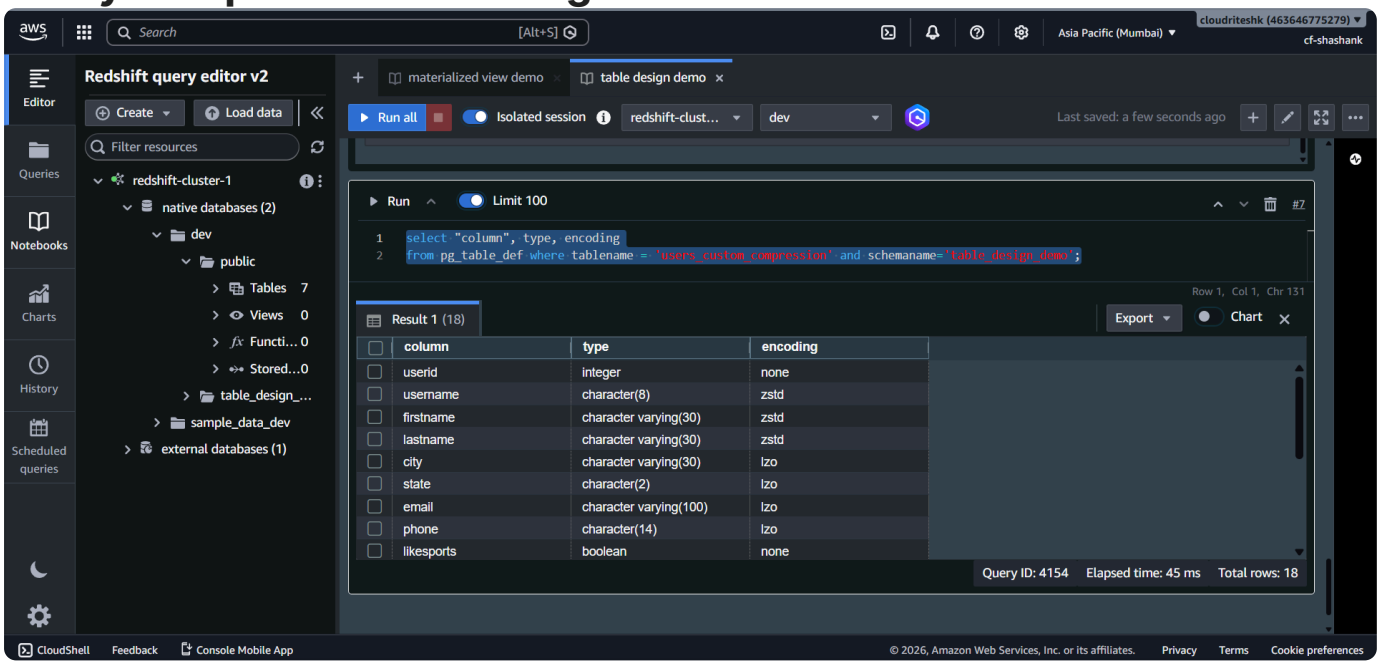
Insert Data




```
INSERT INTO users_custom_encoding
SELECT userid,
       username,
       firstname,
```

```
        lastname,
        city
FROM public.users;
```

Verify Compression Encoding




```
SELECT tablename,s
       "column",
       encoding
FROM pg_table_def
WHERE tablename = 'users_custom_encoding';
```

Observation

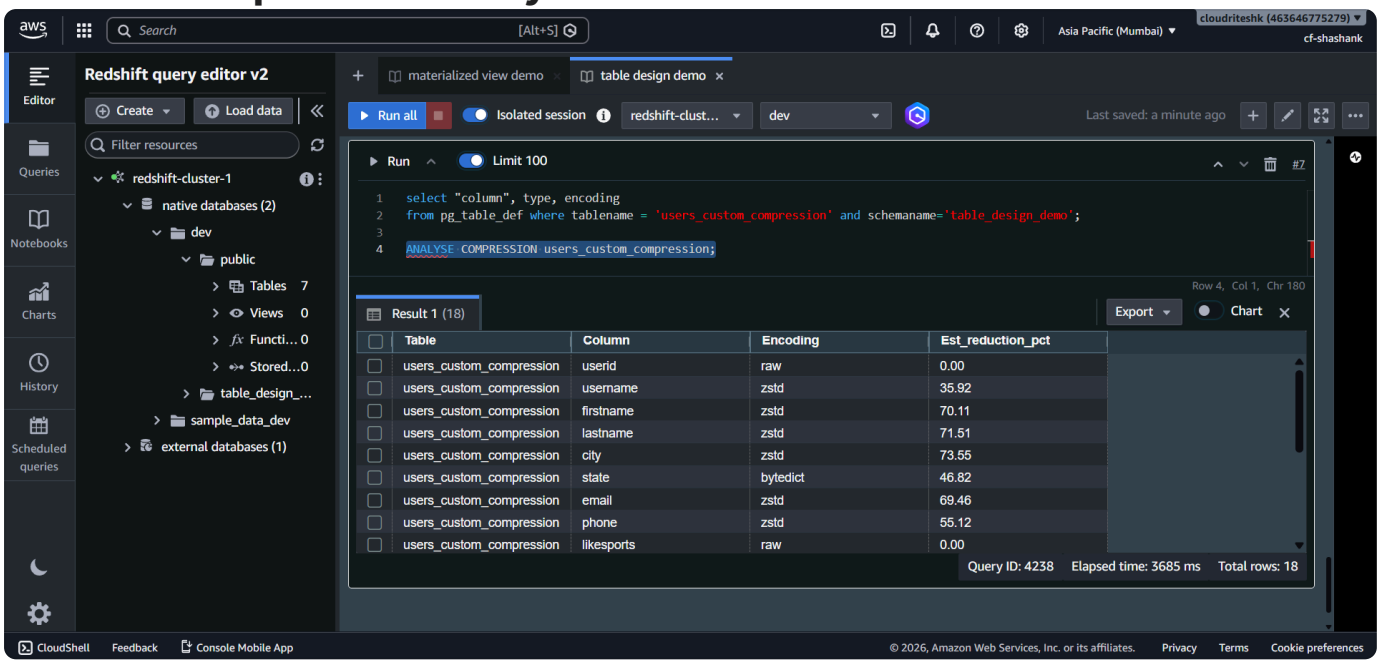
Columns:

- username
- firstname
- lastname

now use: `ZSTD`

Other character columns still use: `LZO` (default Redshift encoding).

Re-Run Compression Analysis




```
ANALYZE COMPRESSION users_custom_encoding;
```

Observation

No recommendations were given for columns already using: `ZSTD`

Only remaining columns received recommendations.

Key Learning

Compression encoding helps:

- Reduce storage usage
- Improve query performance
- Reduce scan cost

Redshift automatically applies compression, but custom encoding can also be manually defined using:

- `CREATE TABLE`
- `ALTER TABLE`
